HABIBA RAFIQUE

M.Sc. Student | HCI & Human-Centered AI Research

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RESEARCH INTERESTS

Human-Computer Interaction, Human-Centered AI, AI for Digital Health, Trustworthy AI, Machine Learning, Large Language Models

EDUCATION

Bangladesh University of Engineering and Technology (BUET) *Dhaka, Bangladesh*
M.Sc. in Computer Science and Engineering Oct 2026 – Present

Bangladesh University of Engineering and Technology (BUET) *Dhaka, Bangladesh*
B.Sc. in Computer Science and Engineering 2022 – 2026

- CGPA: 3.37/4.00 | Final Semester GPA: 3.71/4.00

Holy Cross College *Dhaka, Bangladesh*
Higher Secondary Certificate (HSC) – GPA: 5.00 2018 – 2020

RESEARCH EXPERIENCE

Undergraduate Thesis – Self-Tracking and Digital Health *2025 – 2026*
Department of Computer Science and Engineering, BUET Thesis Completed; Manuscript in Preparation

- **Survey & Data Management:** Designed a 49-item bilingual survey, recruited 533 respondents, and cleaned and encoded data in Python (pandas), handling structural missingness.
- **Statistical Analysis:** Applied Cronbach's α, Spearman correlation, logistic regression, and Random Forest modelling with stratified 5-fold cross-validation and subgroup fairness analysis; visualised findings using Matplotlib and Seaborn.
- **Qualitative Research:** Conducted 9 semi-structured Bangla–English interviews; manually transcribed, translated, and thematically coded responses; analysed open-ended responses using sentence embeddings, PCA, and t-SNE.
- **Ethics & HCI:** Followed informed-consent and anonymisation protocols; translated mixed-methods findings into human-centred recommendations for privacy-by-default, Bangla-first interaction, and compassionate AI feedback.

Hybrid Quantum Machine Learning for Skin Lesion Classification *2025 – 2026*
2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)

- Developed a hybrid quantum-classical pipeline for skin lesion classification using MobileCLIP features, PCA-based dimensionality reduction, and quantum circuit learning.
- Performed comparative evaluation of classical and quantum models using ROC analysis and confusion matrices.

Research Trainee – LLM Alignment Research *Jul 2025 – Jan 2026*
Advanced Machine Intelligence Research Lab (AMIRL) Remote

- Conducted a systematic literature review on large language model alignment techniques, including Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO).
- Compared online and offline alignment approaches in terms of training stability, computational efficiency, and generalisation performance.

Research Internship Program – CHIRAL Bangladesh *Cohort 9.0*
Center for Health Innovation, Research, Action and Learning Bangladesh

- Participated in a structured research internship involving literature review, research gap identification, scientific writing, and research problem formulation.

TECHNICAL SKILLS

Programming Languages: Python, R, Java, JavaScript, TypeScript, C/C++, SQL
Data Analysis & Visualization: Pandas, NumPy, scikit-learn, Matplotlib, Seaborn
ML/DL Frameworks: PyTorch, TensorFlow
Generative AI: RAG, LangChain (Fundamentals), Gemini API, OpenAI API, Hugging Face API, Qdrant
Web Development: React.js, Node.js, Express.js, MongoDB, PostgreSQL, REST APIs
Tools & Platforms: Git, Docker, Linux, VS Code, LaTeX, Figma

SELECTED PROJECTS

AI-Powered Learning Platform with RAG Pipeline – StuddyBuddy *GitHub*

- Developed a RAG-based learning platform using document processing, embeddings, Qdrant, and Google Gemini API for context-aware response generation.
- Deployed the application using Docker, Azure VM, and GitHub Actions CI/CD.

ByteFixers – Project Management Tool *GitHub*

MERN, Gemini AI API, Cloudinary

- Developed a collaborative project management system with AI-assisted features, JWT authentication, and CI/CD workflow.

TourHub – Full-Stack Tour Booking Platform *GitHub*

HTML, CSS, JavaScript, Node.js, Oracle DB, Bootstrap

- Developed a full-stack platform with tour search, booking, payment, and administrative features.

ACHIEVEMENTS & CERTIFICATIONS

- Secured **1st Runner-Up** position at the Eco Tech Hackathon 2026, organized by BUET.
- *Supervised Machine Learning: Regression and Classification* – Coursera (Certificate).
- Participant, Learnathon 3.0: Fresher Training Program by Geeky Solutions (Certificate).
- Solved 150+ SQL problems on LeetCode and HackerRank.

LANGUAGES

English (Fluent) | Bengali (Native)